Middle East Nuclear Development Pressure Forecast

Iran-Israel-Gulf conflict | Conditional OSINT assessment | Evidence through 25 August 2026

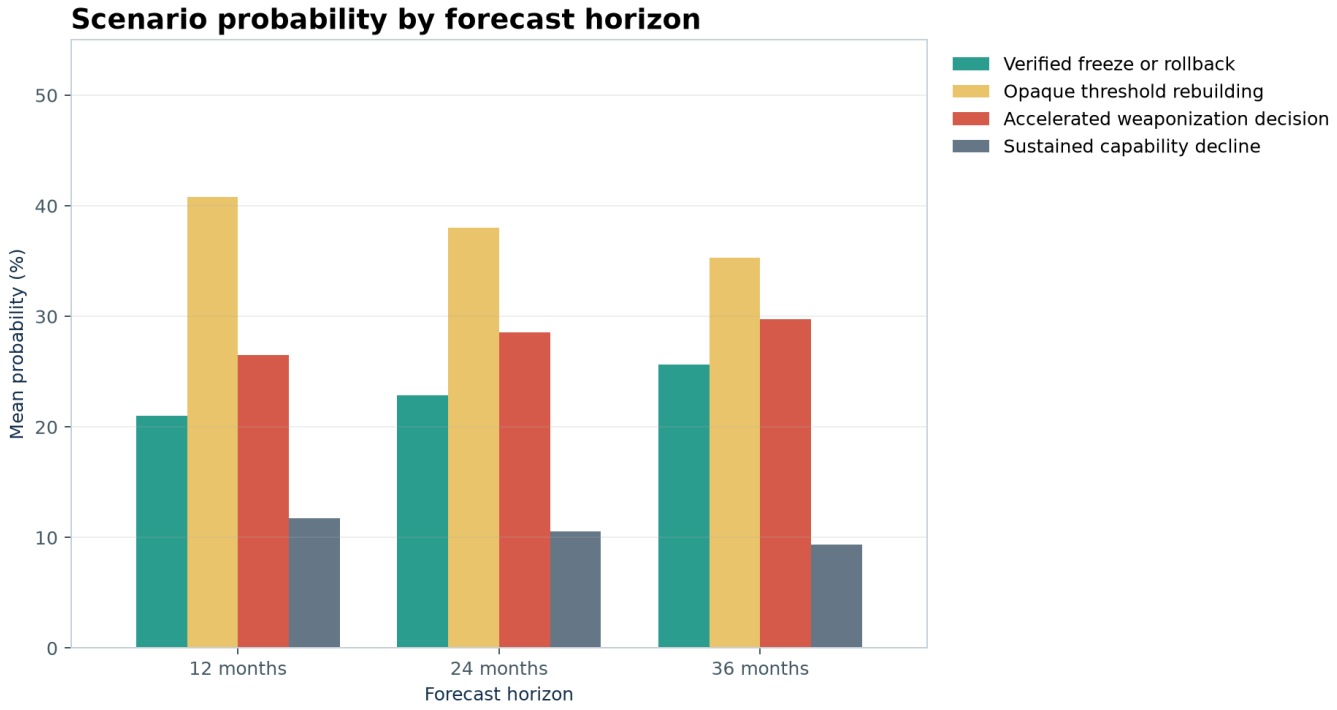
The current evidence produces a 67.3% mean 12-month probability of a push toward greater nuclear latency or a weaponization decision. The leading pathway is *opaque threshold rebuilding* at 40.8%.

Key judgment

Conflict-driven motivation is rising faster than verified technical capability, while observability has deteriorated. This creates a high-pressure but highly uncertain environment: military damage can delay technical progress at the same time that attacks strengthen the perceived value of a strategic deterrent.

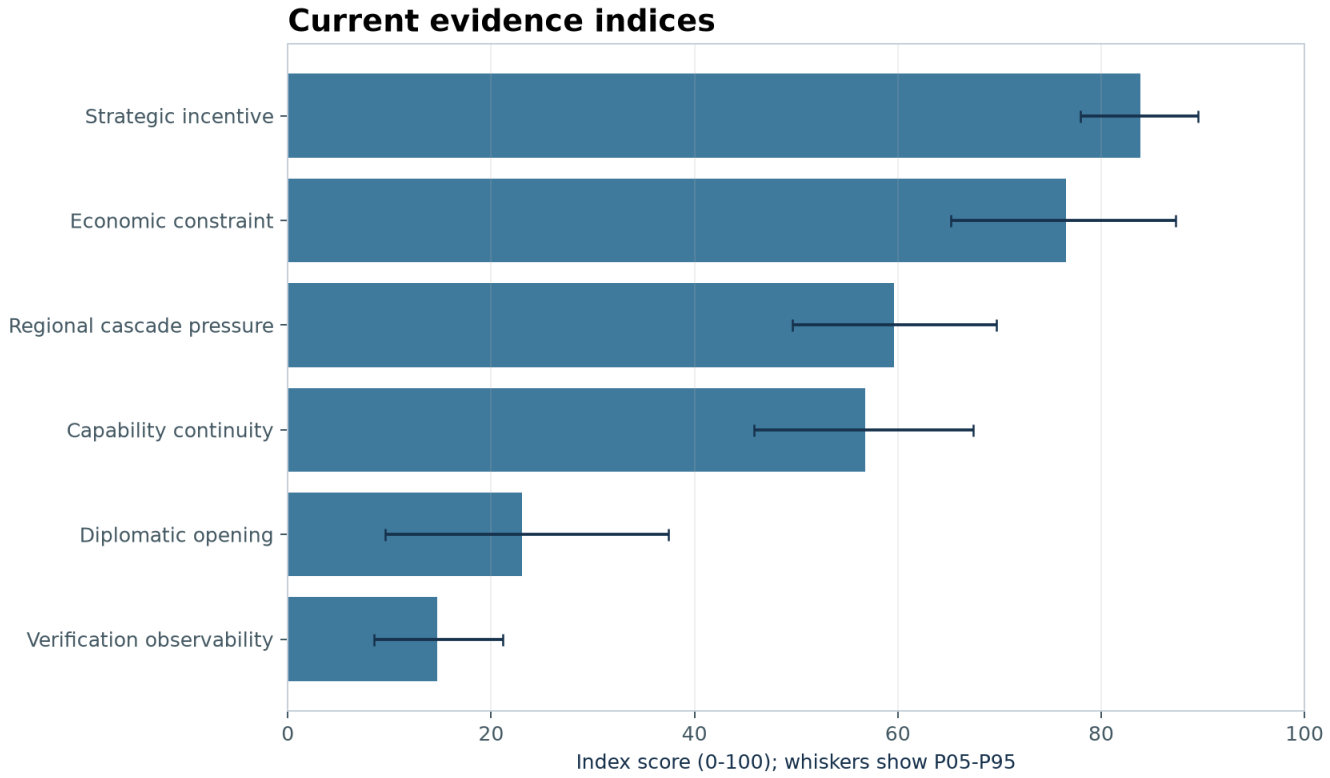
What the model does - and does not do

The model estimates strategic pathways from public evidence. It does not claim that Iran has decided to build a weapon, does not estimate weapon design details, and does not treat missing inspections as proof of weaponization. Low observability widens uncertainty and shifts weight toward opaque pathways.



Simulation settings: 10,000 Monte Carlo draws; reproducible seed 20260821. Probabilities are preliminary analytic estimates, not observed frequencies.

Current evidence structure

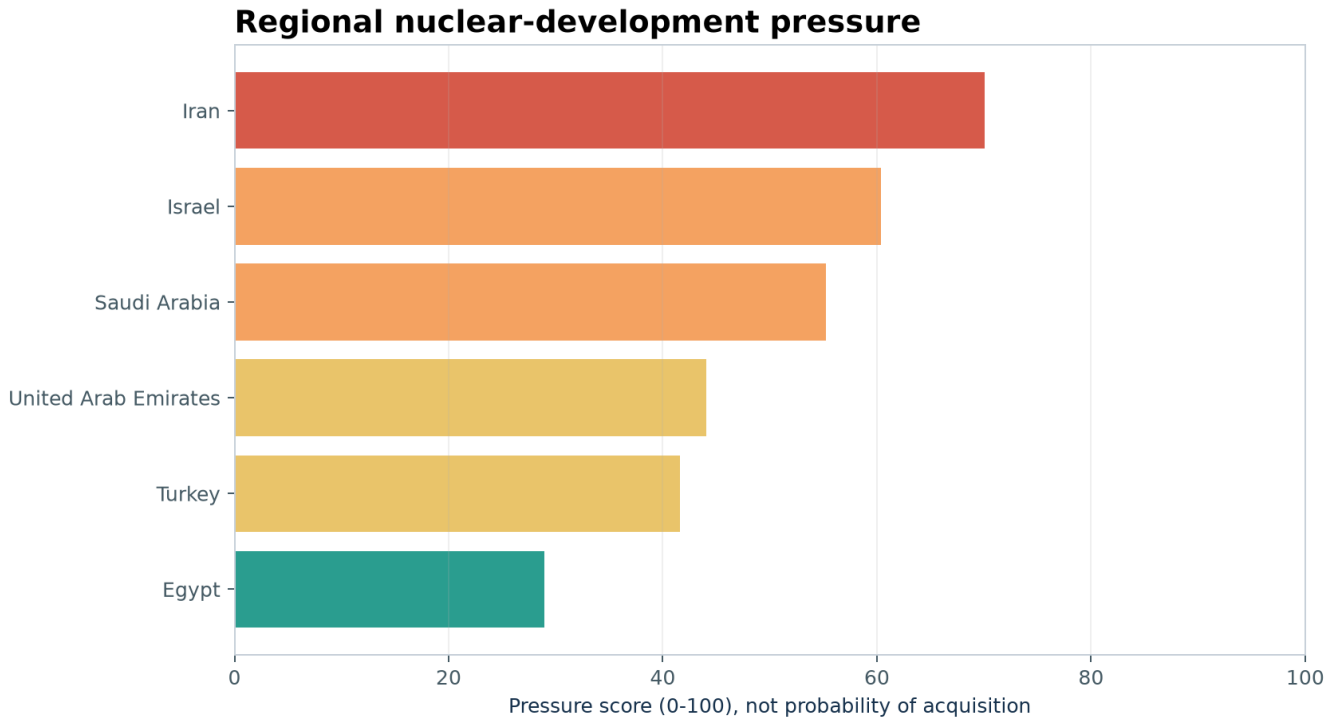


Index	Mean	P05	P95
Strategic incentive	83.9	77.9	89.5
Capability continuity	56.8	45.9	67.4
Verification observability	14.7	8.5	21.2
Diplomatic opening	23.0	9.5	37.4
Economic constraint	76.6	65.2	87.3
Regional cascade pressure	59.6	49.6	69.7

Interpretation: strategic incentive and economic constraint can both be high. The former increases demand for deterrence; the latter reduces implementation capacity and can increase the attraction of negotiations. Verification observability is a confidence-control variable, not a direct measure of prohibited activity.

Regional cascade assessment

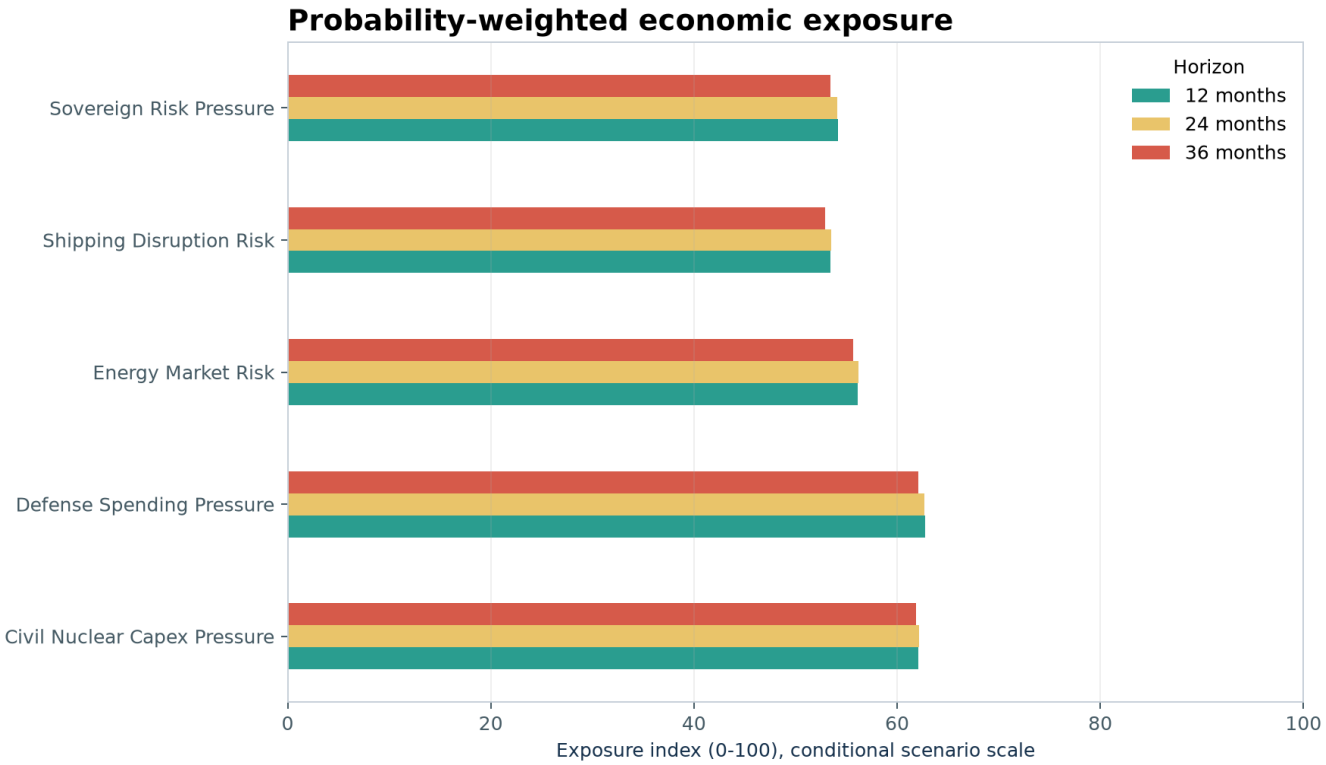
The regional score measures pressure to expand hedging, latency, or strategic nuclear options. It is not the probability that a state will acquire a weapon.



Actor	Pressure	Band	Analytic role
Iran	70.0	High	Primary actor; pressure reflects latency and weaponization pathways under direct attack and low observability
Israel	60.4	Elevated	Assessed nuclear-armed state; score represents modernization and posture pressure rather than new acquisition
Saudi Arabia	55.3	Elevated	Highest secondary hedging pressure; civilian expansion is not evidence of weaponization
United Arab Emirates	44.1	Guarded	High threat exposure but strong safeguards and nonproliferation commitments restrain weapons pathways
Turkey	41.6	Guarded	NATO ties restrain acquisition pressure while regional autonomy and civilian capacity support hedging
Egypt	29.0	Low	Strategic status concerns exist but fiscal and technical constraints limit near-term options

Economic transmission layer

This layer connects the strategic forecast to energy, shipping, defense expenditure, sovereign risk, and civil nuclear investment. Values are conditional exposure indices designed to interface with a separate market model; they are not price forecasts.



Horizon	Exposure metric	Index
12 months	Energy Market Risk	56.1
12 months	Shipping Disruption Risk	53.5
12 months	Defense Spending Pressure	62.7
12 months	Sovereign Risk Pressure	54.2
12 months	Civil Nuclear Capex Pressure	62.1
24 months	Energy Market Risk	56.2
24 months	Shipping Disruption Risk	53.5
24 months	Defense Spending Pressure	62.7
24 months	Sovereign Risk Pressure	54.1
24 months	Civil Nuclear Capex Pressure	62.2
36 months	Energy Market Risk	55.7
36 months	Shipping Disruption Risk	52.9
36 months	Defense Spending Pressure	62.1
36 months	Sovereign Risk Pressure	53.5
36 months	Civil Nuclear Capex Pressure	61.8

Sensitivity and monitoring

Largest modeled changes to the 12-month push probability from a 10-point indicator shock:

Indicator shock	Push probability	Change
Economic Pressure (+10)	62.7%	-4.7 pp
Direct Attack Exposure (-10)	62.7%	-4.6 pp
Iaea Access (+10)	62.8%	-4.5 pp
Economic Pressure (-10)	71.3%	+4.0 pp
Fissile Material Continuity (-10)	63.6%	-3.7 pp
Regional Security Anxiety (-10)	63.8%	-3.5 pp
Diplomatic Breakdown (-10)	63.9%	-3.4 pp
Iaea Access (-10)	70.6%	+3.3 pp
Regional Security Anxiety (+10)	70.1%	+2.8 pp
Facility Survivability (-10)	64.8%	-2.5 pp

Scenario probability table

12-month pathway	Mean	P05-P95
Verified freeze or rollback	21.0%	3.7-50.3%
Opaque threshold rebuilding	40.8%	21.1-51.8%
Accelerated weaponization decision	26.5%	7.8-46.2%
Sustained capability decline	11.7%	2.3-28.1%

Priority indicators for updating

1. Restored IAEA access and verified material accountancy.
2. Evidence of a durable nuclear agreement with enforceable monitoring.
3. Confirmed facility reconstruction or continuing enrichment.
4. Leadership statements that alter official nuclear doctrine.
5. Saudi, Turkish, Egyptian, or UAE changes to safeguards, enrichment policy, or security guarantees.
6. Changes in conflict intensity, sanctions, oil exports, or regime stability.

Confidence assessment

Direction of strategic incentive: moderate confidence. Technical capability continuity: low confidence. Regional cascade ordering: low-to-moderate confidence. Overall scenario probabilities remain preliminary because the conflict regime is new and the frozen forecasts have not matured.

Validation and update governance

Version 0.2 separates validation of model behavior from measurement of forecast performance. It freezes issued probabilities before outcomes are known, preventing a hindsight backtest from being presented as calibration evidence.

Control	Result	Interpretation
Structural validation	4/4 passed	Synthetic edge cases; not a historical backtest
Source freshness	9 current; 0 overdue	Limits vary by source role
Forecast registry	0 scored; 3 open	Scoring begins only after target dates and adjudication
Evidence updates	0 applied	Accepted updates are confidence-discounted
v0.1 baseline drift	0.000 pp maximum	No evidence update means only Monte Carlo rounding drift

Structural validation cases

Case	Expected	Dominant	Result
SV01	Verified freeze or rollback	Verified freeze or rollback	Pass
SV02	Opaque threshold rebuilding	Opaque threshold rebuilding	Pass
SV03	Accelerated weaponization decision	Accelerated weaponization decision	Pass
SV04	Sustained capability decline	Sustained capability decline	Pass

Frozen forecast registry

Horizon	Target date	Status
12 months	2027-08-21	open not matured
24 months	2028-08-21	open not matured
36 months	2029-08-21	open not matured

Brier score and log loss are intentionally blank until a forecast target date passes and the realized scenario is independently adjudicated. This is a feature of the governance design, not missing analysis.

Methodology and source notes

The engine applies any accepted, dated evidence adjustments to the preserved baseline; confidence discounts the adjustment magnitude. It then samples indicators from bounded normal distributions, calculates six weighted indices, and converts standardized deviations into mutually exclusive scenario probabilities using a softmax transformation. Scenario priors are set independently for 12, 24, and 36 months. Regional actors use a separate incentive-capability-restraint model. All values remain editable in CSV files with audit outputs.

Primary source anchors

IAEA GOV/2026/33 (4 June 2026)
IAEA Director General statement (8 June 2026)
UN Security Council Iran nuclear coverage (9 June 2026)
SIPRI Yearbook 2026 release (8 June 2026)
IAEA safeguards framework

Current conflict context

Reuters: Iran sanctions and economic pressure (21 August 2026)
Reuters: conflict posture and stalled diplomacy (17 August 2026)

Prepared as an open-source strategic decision-support product. It is not a statement of government intent, not a substitute for classified intelligence, and not investment advice.